

TAREK H. EL-TAHAWY

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ASSISTANT PROFESSOR, CIVIL ENGINEERING WATER RESOURCES

More than twenty-year professional career in Water Resources engineering specializing in Fluid Mechanics, Hydraulics, Hydrology, Pipe network, Fluvial Hydraulics and River Engineering. Progressive responsible experience is in the field of **teaching** and **research** activities as well as industrial experience in the design of open channels, highway drainage systems, detention/retention basins, hydraulic structures, scour analysis and floodplain delineation. Familiarity with many, general and special, software such as River Analysis System (HEC-RAS), HEC-GeoRAS, HEC-1, DDMSW, CulvertMaster/FlowMaster by Haestad Methods, Visual Urban for Roadway drainage by FHWA, AutoCAD/Civil 3D, ARCGIS, and advanced knowledge of the Microsoft Office Package, as well as FORTRAN and Visual Basic programming languages.

Skills/Strengths

- Confident, competent, and results-oriented candidate
- Strong cross-cultural communications and liaison experience throughout Middle East, Canada and USA
- Proven analytical and problem solving skills
- Excellent technical writing skills

Education

- 2004 Ph.D. Civil Engineering (Hydro-technical), Queen's University, Kingston, Ontario, Canada.
- 1997 M.Sc. Civil Engineering (Water Resources), Helwan University, Cairo, Egypt.
- 1992 B.Sc. Civil Engineering, Helwan University, Cairo, Egypt.

Training/Seminars

- 2008 Natural and Cultural Resource Regulatory Compliance, *Section 404 Clean Water Act and Section 106 National Historic Preservation Act* Consultant Workshop, City of Phoenix, Arizona.
- 2007 Hydrologic Modeling Guidelines (SS 10-07), City of Chandler Arizona
- 2007 Drainage Design Management System for Windows (DDMSW 3.3.2), Flood Control District of Maricopa County (FCDMC) Seminar.

Residency/Visa Status

Canadian

Honors and Awards

- **Best Teaching Assistant Award**, Queen's University, Winter 2004
- Ontario Graduate Scholarship (**OGS, \$15,000**), Queen's University, 2003-2004
- Ontario Graduate Scholarship (**OGS, \$15,000**), Queen's University, 2002-2003
- Ontario Graduate Scholarship for Science and Technology (**OGSST, \$10,000**), Queen's University, 2002-2003 (declined)
- Queen's Graduate Award (**QGA, \$7,500**), Queen's University, 2003-2004
- The Louisa A. Fowler Graduate Fellowship (**\$10,000**), Queen's University, 2001-2002
- Tuition Bursary, Queen's University, 2001-2002.
- Queen's Graduate Fellowship (**QGF, \$7,500**), Queen's University, 2000-2001
- Egyptian Ministry of Higher Education Excellence Award (**HEEA**), Egypt 1991-1992

Professional Membership and Activities

- Registered Professional Engineer (**PE**), Board of Technical Registration (BTR), Arizona, USA. RN: 51562.
- Member of the International Association of Hydraulic Engineering and Research (IAHR).
- Reviewer of the Journal of Hydraulic Engineering, American Society of Civil Engineers.
- Reviewer of the Journal of Hydraulic Engineering, Canadian Society for Civil Engineering.
- Certified Engineer, Egyptian Syndicate of Engineers, Egypt.

Employment Record

- **Assistant Professor**, *Taif University*, AL-Taif, Saudi Arabia, (Sept 2014 to Present), Lectured Fluid Mechanics, Hydraulics, Environmental Hydraulics, Sanitary Drainage, and Irrigation courses. Refereed senior capstone projects for Water Resources and Wastewater Engineering. Participated in department meetings and seminars. Elected as chairman of the **Capstone Project Committee (CPC)**.
- **Water Resources Engineer**, *Primatech Engineers & Consultants*, Phoenix, Arizona, USA (2007 to 2014). Responsibilities and duties included:
 - Project planning, scheduling, allocation of personnel and resources, budgeting, and regular communication with clients, stakeholders, and the public for diverse Water Resource Engineering projects.
 - Project coordination, technical leadership, and technical design for public and private sector clients in Arizona.
 - Administered project proposals, work plans, and project costing/budgeting.

- Performed hydrologic and hydraulic modeling for storm water management using software such as HEC-1, HEC-RAS, HydraFlow, XPSWMM, and others.
- Performed design of channels, culverts, ponds, Energy dissipators, and other hydraulic structures, erosion and scour protection, and sediment control.
- Designed roadway *offsite Drainage Systems* such as CMP's, CBC's, and Open channels, etc. and *onsite Drainage Systems* such as Inlets, Curb openings, Scuppers, Downdrains/Spillways, Side ditches, Retention/Detention basins, etc.
- Attended weekly meetings and/or conference call meetings to discuss projects progress and outstanding issues.
- Addresses and resolved comments from projects' reviewers, particularly from the Flood Control District of Maricopa County (**FCDMC**), Arizona Department of Transportation (**ADOT**), City of Scottsdale and City of Phoenix.
- Prepared technical reports and presentations.
- Contributed to business and client relationship development.

Some of the duties, projects worked on include:

- **Project Manager, Broadway Road Sewer Main Replacement Project, City of Phoenix, Arizona** – Managed, and supervised the replacement of the existing sanitary sewer main along Broadway Road corridor with a new, larger main with increased capacity to meet future growth. The overall project has been broken into four segments, namely, 2A, 2B, 2C, and 2D. This project has the following major benefit:

- New sewer main designed to accommodate flow projections through year 2055.
- New main would relieve existing and predicted Broadway Road capacity issues.
- Odor issues should be improved with higher volumes of wastewater flowing faster through the new main on Broadway Road and surrounding neighborhoods.
- Improved integrity of the Broadway Road sewer main.

- **Project Manager, Upper Camelback Wash, City of Scottsdale, Arizona** – Led bi-weekly meetings to discuss the progress of the 14 million dollar Upper Camelback Wash (UCW) Project. Supervised a team of design engineers and technicians. Participated in public meetings and open houses exhibits. This project included analysis of the existing condition of the wash and recommendations for improvements to the UCW to reduce the incidence of flooding and, if possible, convey the 100-year discharge.

- **Project Engineer, US93 Hoover Dam to MP17** – Completed data collection and review and field investigation to gather necessary information to incorporate into the design. Performed on-site hydrology analysis using the Rational method to calculate the discharges at selected locations for the necessary downdrains/spillways. Completed hydraulic analysis using HY8 for 35 culverts crossing the roadway in the project area. Completed the 50-year and 100-year design storms hydraulic calculations to determine dike relocations, ditches alignment, culverts and channels. Sized the culverts using ADOT Standards.

Prepared and calculated erosion protection for culvert's outlets and embankment. Completed the drainage design memorandum and construction documents.

- Design Engineer US70, Salt Creek Culvert Emergency Repair, Arizona Department of Transportation – This project involves hydrology, hydraulic analysis, and energy dissipation basin/outlet erosion protection design. The design was complicated due to the project features related to an emergency repair, and a limited project budget. Tasks performed included review and evaluation of various energy dissipation basins and outlet protection measurements and submittal of two types of outlet protection alternatives that were applicable to the project site. The proposed alternatives incorporated a gabion drop structure with apron system and a rigid boundary basin. Concept design exhibits as well as hydraulics modeling and calculations were completed and documented in the drainage memorandum.

- Project Engineer, I-17/Cordes Junction T.I. – Performed hydrologic and hydraulic analyses and designs for the drainage system (on-site and offsite) required to collect and convey storm water runoff to the discharge area. The work consisted of data collection, review and investigation. For the culverts, the hydraulic analysis was performed for the 50-year and 100-year design storms and culverts were sized according to ADOT standards using the FEMA-approved CulvertMaster software for the existing and proposed culverts crossing the roadway. Designed the pavement drainage system including inlets, curbs openings, scuppers/downdrains, and side ditches. Performed erosion protection calculations, prepared the drainage report and construction documents.

- Design Engineer, Riggs Road Drainage Design, City of Queen Creek, Arizona – This project includes review, analysis, and design of the on-site drainage system for a 3-mile segment of Riggs Road from Ellsworth Road to Meridian Road. The drainage design for this project includes hydraulic analysis and design for the on-site pavement drainage system. Basically, the drainage system consists of retention basins located along both the north and south sides of the roadway to capture pavement flow. At the westernmost end of the project and due to the roadway super-elevation, a catch basin system that drains to the south side of the road was adopted. In addition a split flow analysis was performed at the intersection of Riggs road with the east branch of Sonoqui Wash at Crismon Road intersection.

- **Assistant Professor, Helwan University, Cairo, Egypt, (2005 to 2006).** Lectured Fluid Mechanics, Hydraulics and surface water Hydrology courses and supervised two M.Sc. students.
- **Teaching/Research Assistant, Queen's University, Kingston, Ontario, Canada (2000 to 2004).** Assisted with class and laboratory teaching and marked assignments/reports for Fluid Mechanics and Hydraulics courses. Assisted four post-graduate students on their research projects.
- **Teaching/Research Assistant, University of Windsor, Windsor, Ontario, (1999 to 2000).** Assisted with teaching and lecturing the Hydraulics, Fluid mechanics, and water treatment courses.
- **Assistant Lecturer, Helwan University, Cairo, Egypt, (1998 to 1999).** Taught and lectured the Hydraulics and Irrigation, Fluid mechanics, Civil drawing, Structure and Surveying courses.
- **Project Management Engineer, African Consultant Office (ACO), (1995-1999),** fully responsible for developing time scheduling programs of construction projects,

optimizing resources, prioritizing work, follow-ups, cost control, and contractors' administration.

- **Lab. Engineer**, *Helwan University*, Egypt, (1993 to 1998). Led lab sessions, prepared undergraduate experimental course materials, wrote lab manuals/software, worked on setting up experiments, and analyzing the results.

Extracurricular Activities

Multimedia teaching

- Excellent use of computers and multimedia in classroom.
- Built and customized course materials in a progressive and enjoyable teaching strategy.
- Enhanced the overall teaching process by incorporating the lessons to correspond with multimedia interactive presentations.
- Encouraged the students to build their own documents as a part of their assigned work.

Project management

- Project Management senior engineer (PT) (1995-1999), fully responsible for developing time scheduling programs for construction projects, optimizing resources, prioritizing work, follow-ups, cost control, and contractors' administration.

Contributions to Research and Development

- I. ElShinnawy, A. Bestawy, and T. ELTahawy (2017): "Assessment and Management of Flash Floods for Sustainable Development in Al-Sail Al Kabir Area, Kingdom of Saudi Arabia". International Journal of Applied Engineering Research ISSN 0973-4562 Volume 12, No.11 (2017) pp. 2807-2814.
- M. Eizel-Din, M. D. Bui, T. El Tahawy, and P. Rutschmann (2010): "Numerical investigation of grain size effect on bed deformation in meandering channel". Proceeding of the First European Congress of the IAHR, Edinburgh, UK.
- T. El-Tahawy, A.M da Silva, S. Holzwarth and E. Pasche (2008): "Bed topography of alluvial meandering streams under varying width-to-depth ratios". Proceedings of the River Flow 2008, International Conference on Fluvial Hydraulics, Çeşme. İZMİR, TURKEY.
- da Silva, A. M. F, El-Tahawy, T. H. (2008): "On the location in flow plan of erosion-deposition zones in sine-generated meandering streams", Journal of Hydraulic Research, IAHR, Volume 46, p. 49-60.
- da Silva, A. M. F., El-Tahawy, T. H., and Tape, W. D. (2006): "Variation of flow pattern with sinuosity in sine-generated meandering streams". ASCE, Journal of Hydraulic Engineering, Vol. 132, No. 10.
- A.M. Ferreira da Silva and T. El-Tahawy (2006): "Location of hills and deeps in meandering streams: an experimental study". Proceedings of River Flow 2006, International Conference on Fluvial Hydraulics, Lisboa, Portugal.
- El-Tahawy, T. H. (2004): "Patterns of flow and bed deformation in meandering streams: An experimental study". Ph.D. Thesis, Department of Civil Engineering, Queen's University, Kingston, Ontario, Canada.

- da Silva, A. M. F., Yalin, M. S., El-Tahawy, T. H. and Tape, W. D. (2001): "Patterns of flow in meandering channels of varying sinuosity". Proceedings of the XXIX Congress of the International Association of Hydraulic Engineering and Research (IAHR), Beijing, China, September 16-21.
- da Silva, A. M. F., Tape, W. D. and El-Tahawy, T. H. (2001): "Mechanics of turbulent flow in meandering channels of varying sinuosity: an experimental study". Proceedings of the 29th Annual Conference, Canadian Society for Civil Engineering, CSCE, May 30-June 2.
- El-Tahawy, T. H., Khalil, M. B., and Zein, S. A. (2000): "Study of the characteristics of diaphragm orifice meters in non-circular conduits". Proceedings of the 2nd International Conference of Civil Engineering, ICCE, Helwan University, Cairo, Egypt.
- Tape, W. D., El-Tahawy, T. H. and da Silva, A. M. F. (2000): "Experimental investigation of flow in meandering channels". Abstract, Proceedings of the 28th Annual Conference, Canadian Society for Civil Engineering, CSCE, June 7-10.
- El-Tahawy, T. H. (1997): "Study of the characteristics of diaphragm orifice meters in non-circular conduits". M.Sc. Thesis, Civil Engineering Department, Helwan University, Cairo, Egypt.

References:

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3. Dr. Magdi El Emam Title: Associate Professor Department of Civil Engineering School of Engineering American University of Sharjah (AUS) Sharjah, P.O. Box 26666 United Arab Emirates Tel: (971) 6-5152948 e-mail: melemam@aus.edu	4. Dr. Tahany Youssef Title: Professor Faculty of Engineering, Mattaria Department of Civil Engineering Helwan University Cairo Egypt Tel: 0226353944 e-mail: tahanyoussef@yahoo.com

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